



Screen media use and sleep disturbance symptom severity in children

Garrett C. Hisler, PhD^{a,*}, Brant P. Hasler, PhD^a, Peter L. Franzen, PhD^a,
Duncan B. Clark, MD, PhD^a, Jean M. Twenge, PhD^b

^a University of Pittsburgh, Pittsburgh, Pennsylvania, USA

^b San Diego State University, San Diego, California, USA



ARTICLE INFO

Article History:

Received 4 March 2020

Revised 2 July 2020

Accepted 3 July 2020

Keywords:

Sleep duration

Sleep onset latency

Sleep-wake disturbances

Screen media

Electronic device use

ABSTRACT

Objectives: Few studies have sought to evaluate how screen media use relates to symptoms of sleep-wake disturbances. To extend these prior studies in a large sample of children, this study examined associations of different types of screen media with symptom severity of different classes of sleep-wake disturbances. This study was preregistered here.

Design: This study utilized the baseline cross-sectional survey administered within the Adolescent Brain Cognitive Development Study (ABCD; Release 2.0).

Participants: ABCD recruited over 11,000 U.S. children age 9–10 across 21 study sites using an epidemiologically-informed school-based recruitment strategy.

Measurements: Children reported typical weekend and weekday use of TV, video, video game, social media, texting, and video chat, and parents completed reports of the child's symptom severity of sleep-wake disturbances via the Sleep Disturbance Scale for Children.

Results: Greater screen media use, TV, video, and video game use, was associated with decreased sleep duration, increased sleep onset latency as well as greater excessive sleepiness, insomnia, and overall sleep disturbance symptom severity. Use of these screen medias were also associated with clinically relevant sleep problems. Ethnoracial differences emerged in screen use and sleep, but did not moderate the association between screen use and sleep.

Conclusions: Greater use of screen medias was not just associated with longer sleep onset latency and shorter sleep duration, but also increased severity of multiple types of sleep-wake disturbances. Future research should use longitudinal designs to determine the direction of these associations in adolescent populations.

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Children who spend too much time using screen media devices are more likely to have sleep problems.^{1–3} These associations can arise because too much time on screen media may displace sleep, the blue light emitted from screens may delay melatonin production and delay sleep, and content on screen media may be arousing and inhibit sleep.^{4,5} These detrimental effects underlie pediatric recommendations that screen time should be limited to less than two hours in children and children.⁶ To date, existing evidence has primarily focused on the associations between various screen media and common characteristics of sleep, such as sleep duration and sleep onset latency. Despite a multitude of evidence that increased screen time is associated with worse sleep, few studies have examined if screen media use relates to sleep-wake disturbances (e.g., insomnia, hypersomnolence). To advance the understanding of how screen media is related to sleep, this study utilized the Adolescent Brain Cognitive

Development (ABCD: abcdstudy.org) study to examine in a large sample of children whether greater use of different types of screen media are related to the symptom severity of different classes of sleep disturbances that occur both during wakefulness (e.g., long sleep onset, excessive daytime sleepiness) and during sleep (e.g., arousals from sleep, sleep disordered breathing), which are collectively referred to as “sleep-wake disturbances” in this study.

Screen time and sleep-wake disturbances

Thus far, prior studies have largely focused on how parent reports of child's TV use and sleep-wake disturbances are associated. Specifically, more frequent TV use has been linked to greater insomnia symptoms, sleep disordered breathing, sleep-wake transition disturbances (i.e., parasomnias that occur during transitions in sleep), and hypersomnolence concerns in children and children.^{3,7–10} Only one prior study has examined whether other kinds of screen media uses besides TV use are associated with sleep-wake disturbances.³ This study found that time spent on a computer did not have any

*Corresponding author: Garrett C. Hisler, Department of Psychiatry, University of Pittsburgh, Pittsburgh, Pennsylvania 15260, USA.

E-mail address: hislergc@upmc.edu (G.C. Hisler).

associations with symptoms of a broad range of sleep-wake disturbances, although more frequent computer use did relate to more normative sleep characteristics such as later sleep-wake times and shorter sleep duration. Thus, while prior evidence supports that TV use is related to sleep-wake disturbances, it is unclear whether these associations replicate with other screen media types (such as gaming or social media use).

It is important to note that these studies tended to report effect sizes of approximately $d = 0.20$. Such effect sizes are frequently interpreted as small; however, it is important to consider that not only is sleep a daily behavior that carries profound economic, mental and physical health consequences, but also that children represent a population in which sleep is particularly vulnerable and critical for health and in which both screen time and sleep are amenable to intervention.^{11–13} Thus, despite seemingly small associations, the relation between screen use and sleep may be one potentially actionable area to improve sleep disturbances in children. Unfortunately, most of the prior studies focusing on sleep-wake disturbances are over a decade old and the kinds and prevalence of different screen medias have greatly evolved (e.g., greater video game and mobile device use, using social media on mobile devices).^{14,15} Importantly, prior studies assessing more normative sleep characteristics (e.g., sleep duration, sleep timing, sleep onset latency) have found differential associations between different kinds of screen media uses and normative sleep characteristics.^{15,16} Thus, there is a need to understand how newer and different types of screen media are associated with sleep-wake disturbances.

Screen media devices can differ along a variety of dimensions that are relevant to consider in relation to sleep, such as portability and interactivity. More portable and interactive devices tend to be used in bed at nighttime and are held closer to the face where blue light can more strongly suppress melatonin secretion, thereby delaying and disturbing sleep.^{5,16} Prior studies found that time spent on portable and interactive screen media devices was more strongly associated with shorter and less sufficient sleep in children than nonportable or noninteractive devices.^{15,16} Because prior research has largely focused on (1) how greater TV use is associated with worse sleep-wake disturbances, or (2) how different screen media uses related to normative sleep characteristics such as sleep duration, it is unknown how different kinds of screen media devices are associated with different kinds of sleep-wake disturbances (e.g., insomnia symptoms, arousal disturbances, sleep disordered breathing).

To address this gap, this study used data from ~11,000 children to examine whether child reported time on various types of screen media was related to parent reports of that child's symptoms of broad classes of sleep-wake disturbances. Study analyses and hypotheses were preregistered here prior to analysis of data. Based upon prior research linking greater TV use to worse sleep-wake disturbances, as well as findings that screen media types that are more portable and interactive are more strongly associated with worse sleep characteristics, the following relationships were hypothesized:

- H1. Greater screen media use will be associated with shorter sleep duration
- H2. Greater screen media use will be associated with longer sleep latency.
- H3. Greater screen media use will be associated with greater symptom severity of insomnia.
- H4. Greater screen media use will be associated with greater symptom severity of excessive sleepiness.
- H5. Greater screen media use will be associated with greater symptom severity of sleep-wake transition disturbances.
- H6. Greater screen media use will be associated with greater total scores of symptom severity of sleep-wake disturbances.

H7. Compared to TV screen media uses (i.e., watching TV and videos), non-TV screen media (i.e., social media use, texting, gaming, video chat) will generally be more strongly associated with a.) sleep-wake disturbances, b.) shorter sleep duration, and c.) longer sleep onset latency.

Screen media use was not hypothesized to be associated with sleep disordered breathing, arousal disturbances, or sleep hyperhidrosis symptom severity because prior studies suggest null associations^{3,7–10}.

Beyond these specific pre-registered hypotheses, the role of ethnoracial group in screen use and sleep was also explored. Prior work suggests that children who belong to ethnoracial minority groups may engage in greater screen use and have worse sleep. For instance, in comparison to white children, Black children are more likely to sleep for shorter periods of time, are less likely to get sufficient sleep, and are more likely to have screen media devices in their bedrooms.^{17–19} Despite such differences in sleep and screen media between White children and children from minority backgrounds, there is a dearth of research on if there are differential relations between screen media use and sleep across ethnoracial groups or differences in sleep may be accounted for by differences screen media use, possibilities which are explored in this study.

Methods

Participants

The ABCD study is a multi-site, longitudinal study that has currently completed its first wave of data collection on 9- to 10-year-old youths. The ABCD study conducted these assessments throughout the year at university and research institution settings with MRI facilities. The ABCD study team used an epidemiologically-informed school-based recruitment strategy that was designed to approximate the demographics of the 21 ABCD sites and the United States as a whole.^{20,21} ABCD research sites either used a central Institutional Review Board at the University of California, San Diego or used a local Review Board for the ethical review and approval of the research protocol.²² We used ABCD Release 2.0 which contains data on 11,878 participants. Listwise deletion was used to handle missing data and therefore sample size ranged from 10,666 to 10,676 depending on variables analyzed. Children in the sample were, on average, 9.91 years old ($SD = 0.62$ years) and 52% of these children were female, 53% identified as White, 20% as Hispanic/Latinx, 18% as Black, 5% as Asian, and 2% as Indigenous American. Children tended to belong to middle to upper class families (Mean parent years of education = 16.61; Mean family income = \$103,309; 69% with a currently employed parent).

Measures

Screen media use

Participants were asked 12 questions about their time on various screen media, with 6 questions referring to use “on a typical weekday” and the other six referring to use “on a typical weekend.” They were asked how many hours they: “watch TV shows or movies,” “watch videos (such as YouTube)?”, “Play video games on a computer, console, phone, or other device (Xbox, Play Station, iPad)?”, “Text on a cell phone, tablet, or computer (e.g., GChat, Whatsapp, etc.)?”, “visit social networking sites like Facebook, Twitter, Instagram, etc.?” and “video chat (Skype, Facetime, etc.)?” Response choices were: “none” (0), “<30 minutes” (0.25), “30 minutes” (0.5), “1 hour” (1), “2 hours” (2), “3 hours” (3), and “4+ hours” (4). These questions were created by ABCD personnel for the study but were

based on measures developed by Sharif and colleagues.²³ Note that the original preregistered analyses plan focused on using time on the various screen media uses as continuous variables. However, due to data skew for the continuous screen media use variables (see supplemental table 1), these variables were dichotomized into 0 (<2 hours of screen time) and 1 ($\geq$ 2 hours of screen time) based upon the pediatric recommendation to limit screen time to less than two hours.⁶

Total screen media use and total non-TV screen media use

In addition to the dichotomous screen media type specific variables, total time spent on each type of screen media were summed together to create continuous indices of total weekday and weekend screen media use. Total time using video games, social media, texting, and video chat were each summed to create an indicator of weekday and weekend non-TV screen media use. Dichotomous versions of each of these variables were also created based upon the aforementioned pediatric recommendations.

Sleep-wake disturbance symptom severity

Participant sleep was evaluated through a parent's report using the Sleep Disturbances Scale for Children.^{24,25} A prior review of sleep measures in children concluded that this scale was “well-established” based upon its psychometric properties, use by multiple investigators, and its ease of access and scoring; this review also concluded that it has good diagnostic accuracy in differentiating between clinical and nonclinical populations.²⁶ This scale contains 26 questions assessing the frequency of symptoms of multiple kinds of sleep-wake disturbances in children. With the exception of the sleep duration and sleep onset latency items (see below), responses are made on a scale from “Never” (1) to “Always (daily)” (5). These summed responses were used as indices of sleep-wake disturbance symptom severity. The following sleep-wake disturbances were assessed: insomnia (seven items, e.g., “the child has difficulty getting to sleep at night”, $\alpha = 0.73$), sleep disordered breathing disturbances (three items, e.g., “the child has difficulty in breathing during the night”, $\alpha = 0.37$), arousal disturbances (three items, e.g., “you have observed the child sleepwalking”, $\alpha = 0.56$), sleep-wake transition disturbances (six items, e.g., “the child startles or jerks parts of the body while falling asleep”, $\alpha = 0.59$), excessive sleepiness (five items, e.g., “the child experiences daytime sleepiness”, $\alpha = 0.72$), and sleep hyperhidrosis (two items, e.g., “the child sweats excessively during the night”, $\alpha = 0.81$). Responses to all items were summed to compute an overall sleep-wake disturbance symptom severity score ($\alpha = 0.65$). Using this overall score, a threshold of 39 was used to identify children with likely clinically significant sleep-wake disturbances.²⁴

Sleep duration

Sleep duration was measured with a single item from the Sleep Disturbances Scale for Children. In this item, parents responded to “How many hours of sleep does your child get on most nights?” Response options were “9–11 hours” (1), “8–9 hours” (2), “7–8 hours” (3), “5–7 hours” (4), and “less than 5 hours” (5). These responses were recoded into the approximate sleep duration each scale point represented (i.e., “9–11 hours” into “10,” “8–9 hours” into “8.5,” “7–8 hours” into “7.5,” “5–7 hours” into “6,” “less than 5 hours” into “2.5”). A dichotomous variable representing insufficient sleep was also calculated based upon the National Sleep Foundations sleep duration recommendations for children in this age group (1 = 7–8 hours of sleep or fewer; 0 = 8–9 hours of sleep or more).²⁷ Although the recommended sleep duration for 9- to 11-year-olds is nine to eleven hours, “8–9 hours” was included in this dichotomization because it included 9 hours.

Sleep onset latency

Sleep onset latency was measured with “How long after going to bed does your child usually fall asleep?” from the Sleep Disturbances Scale for Children. In this item, parents responded to “How long after going to bed does your child usually fall asleep?” Responses options were “less than 15 minutes” (1), “15–30 minutes” (2), “30–45 minutes” (3), “45–60 minutes” (4), and “more than 60 minutes” (5).

Covariates

Age, parent employment (currently working = 1; not currently working = 0), family income (ranging from 1 = less than \$5000 to 10 = \$200,000 and greater), parent education (ranging from 0 = no formal education to 21 = doctoral degree), race (0 = White, 1 = non-White), ethnicity (Hispanic/Latinx = 0, non-Hispanic/non-Latinx = 1), and gender (Male/transgender males = 0; Female/transgender females = 1) were used as covariates in analyses.

Analytic strategy

To examine the association between screen media use and sleep, mean levels of sleep variables were compared between children with more than 2 hours of screen use vs. those with less for the different types of screen medias. Because screen media use can vary between the weekdays when children have school and the weekend when they have less time constraints, these comparisons were conducted separately for weekday and weekend screen use. All comparisons were made after adjustment for the above study covariates. Cohen's *d* effect sizes are presented for comparison of effects across different screen media uses and sleep.

In addition, the clinical significance of the association of screen media use with sleep was examined by examining the risk of having insufficient sleep duration and clinically significant sleep-wake disturbance scores between children with more than 2 hours of screen use vs. those with 2 or less.

Last, the role of ethnoracial group differences were explored by examining differences in the prevalence of and the association between screen use and sleep variables by comparing White children to Black, Asian, Indigenous American, and Hispanic/Latinx children. All study covariates previously mentioned besides the binary race and ethnicity variables are used in these analyses.

In order to limit false-positive findings because of multiple inferential tests, only *P* values under .001 are interpreted. Note that some children within the sample were siblings and twins. Analyses were conducted with and without siblings and twins in the sample to examine the effects of this familial nesting. Findings did not differ when including or excluding siblings and twins; therefore, we present results with the full sample.

Missing data

To examine whether missing data associated with socio-demographic factors may bias results, study covariates were used in binary logistic regressions predicting missingness for weekday screen use, weekend screen and sleep. No study covariate significantly predicted probability of missing data.

Results

Percent of children with 2 or more hours of screen media use are presented in Table 1 (raw mean levels of use are also presented in supplemental Table 1). More children spent two or more hours on TV, video, and video game use than on texting, social media, and video chat use. Engagement in two or more hours of screen media use was greater on the weekend than on weekdays. In terms of sleep, children in the sample had fairly healthy sleep as average sleep

Table 1
Participant characteristics.

Variable	M (SD)
Weekday tv use	26% with ≥ 2 hours of use
Weekend tv use	45% with ≥ 2 hours of use
Weekday video use	20% with ≥ 2 hours of use
Weekend video use	28% with ≥ 2 hours of use
Weekday video game use	21% with ≥ 2 hours of use
Weekend video game use	31% with ≥ 2 hours of use
Weekday texting use	3% with ≥ 2 hours of use
Weekend texting use	4% with ≥ 2 hours of use
Weekday social media use	2% with ≥ 2 hours of use
Weekend social media use	2% with ≥ 2 hours of use
Weekday video chat use	2% with ≥ 2 hours of use
Weekend video chat use	3% with ≥ 2 hours of use
Weekday non-tv screen use	26% with ≥ 2 hours of use
Weekend non-tv screen use	37% with ≥ 2 hours of use
Weekday total screen media use	62% with ≥ 2 hours of use
Weekend total screen media use	78% with ≥ 2 hours of use
Sleep duration	9.00 (1.11) hours
Sleep onset latency	1.93 (0.99)
Insomnia disturbance symptom severity	11.76 (3.75)
Sleep disordered breathing symptom severity	3.77 (1.26)
Arousal disturbance symptom severity	3.44 (0.92)
Sleep-wake transition disturbance symptom severity	8.18 (2.63)
Excessive sleepiness disturbance symptom severity	6.95 (2.44)
Sleep hyperhidrosis disturbance symptom severity	2.44 (1.18)
Total sleep-wake disturbance symptom severity	36.54 (8.24)

duration was 9 hours and symptom severity of each class of sleep-wake disturbance was near the low end of its scale.

Associations between screen media use categories and sleep duration and sleep onset latency

Except for video chat and texting use on weekdays and TV use on weekends, children with two or more hours of screen use of any type of screen media on either weekdays or weekends had shorter sleep durations (average $d = -0.19$, all P 's $< .001$, see [Tables 2 and 3](#)). These findings extended to the prevalence of insufficient sleep (see [Figs. 1 and 2](#)). Logistic regressions revealed that children who spent 2 hours or more watching TV and videos and playing video games were more likely to have insufficient sleep (relative risk range: 1.23–1.60, all P 's $< .001$, see [Table 4](#)). Additional logistic regressions using the continuous non-TV screen media and total screen media variables indicated that each one hour increase in screen media use was associated with an increase in the odds of having insufficient sleep (odds ratios range: 1.08–1.10, all P 's $< .001$). [Fig. 3](#) depicts that the prevalence of insufficient sleep increased linearly with total time spent on screen media. In terms of sleep onset latency, video use, video game use, and overall non-TV screen use on both weekdays and weekends predicted longer sleep onset latency (d 's range from 0.05 to 0.14, all P 's $< .05$). These patterns of associations replicate prior findings and support hypotheses one and two.

Is screen media use related to sleep-wake disturbance symptom severity?

Using TV, video, and video games for 2 or more hours on either weekdays or the weekend was associated with increased severity of insomnia symptoms (d 's range from 0.11 to 0.18, all P 's $< .001$, see [Tables 2 and 3](#)). Two or more hours of video or video game use was also associated with increased symptom severity of excessive sleepiness (d 's range from 0.10 to 0.13, all P 's $< .001$). Interestingly, 2 or more hours of TV, video, and video game use on the weekend, but not the weekday, were also associated with greater sleep hyperhidrosis symptom severity (d 's range from 0.08 to 0.12, all P 's $< .001$). Reflecting these associations with more specific sleep-wake

disturbances, these three screen media uses were associated with overall sleep-wake disturbance symptom severity score (d 's range from 0.10 to 0.16, all P 's $< .001$).

In contrast, 2 or more hours of texting, social media, and video chat use across both weekdays and the weekend had no patterns of associations with the symptom severity of sleep-wake disturbances, though two or more hours of social media use on weekend was associated with greater overall sleep-wake disturbance symptom severity ($d = 0.22$, $P < .001$). Both non-TV screen use and total screen use on weekdays and weekends were associated with insomnia, excessive sleepiness, and total sleep-wake disturbance symptom severity (d 's range from 0.07 to 0.15, all P 's $< .001$).

The sleep-wake disturbance symptom severity findings extended to the prevalence of clinically relevant sleep-wake disturbances (see [Figs. 4 and 5](#)). Logistic regressions revealed that children who engaged in two or more hours of TV, video, and video game use on the weekday or weekend had a greater risk of having disturbed sleep (relative risk range 1.23 to 1.60, all P 's $< .001$, see [Table 5](#)). Additionally, separate logistic regressions using the continuous non-TV screen media and total screen media variables as predictors indicated that each one hour increase in screen media was associated with an increase in the odds of having insufficient sleep (odds ratios range: 1.05 to 1.07, all P 's $< .001$). [Fig. 6](#) depicts that the prevalence of disturbed sleep increases linearly with overall time spent on screen media.

Which screen media types have the strongest relations with sleep?

To determine which screen media types had the strongest associations with sleep, the average of the absolute values of the Cohen's d effect sizes for each screen media use variable with each sleep variable was calculated (see rightmost column of [Tables 2 and 3](#)). Three important patterns emerge from these effect sizes. First, the effect sizes were small, but varied in magnitude (range = 0.07–0.14). Second, effect sizes were slightly larger for the weekend screen media use variables than the weekday screen media use variables, though these effects were not necessarily statistically significant when using a P value of .001. Third, TV use had the smallest association with sleep as its average absolute effect size was 0.07 for both weekday and weekend use, whereas social media use had the largest effect sizes (i.e., 0.12 for weekday use and 0.14 for weekend use). However, the non-TV total screen use and total screen use variables had similar sized associations with sleep variables. Overall, these patterns did not support Hypothesis 7 given that non-TV screen media uses (i.e., social media use, texting, gaming, video chat) did not clearly have larger effect sizes than screen-based video uses (i.e., TV use and video use) and specific types of non-TV screen uses tended to not have statistically significant associations with sleep.

Are there ethnoracial group differences in screen use and sleep?

Mean levels of screen use and sleep variables for the different ethnoracial groups are presented in [Table 6](#). In terms of screen use, the only consistent difference to emerge was that more Black children engaged in two or more hours of all screen media uses than White children (all P 's $< .001$). Black children also had shorter sleep durations, greater sleep disordered breathing, and greater sleep-wake transition disturbances than White children (all P 's $< .001$). In contrast, Hispanic/Latinx children had longer sleep durations, shorter sleep onset latencies, and less insomnia symptom severity than White children (all P 's $< .001$). Differences in mean sleep variables remained in subsequent analyses that controlled for differences in total screen use, suggesting that differences in screen use between ethnoracial groups did not account for mean differences in sleep.

Because differences in screen use and sleep emerged between White and Black children and between White and Hispanic/Latinx

Table 2

Association between sleep-wake disturbances and time use of screen medias on weekdays.

Predictor	Sleep duration	Sleep onset latency	Insomnia	Sleep disordered breathing	Arousal disturbances	Sleep-wake transition disturbances	Excessive sleepiness	Sleep hyperhidrosis	Total sleep-wake disturbances	Average absolute <i>d</i>
	M (SE)	M (SE)	M (SE)	M (SE)	M (SE)	M (SE)	M (SE)	M (SE)	M (SE)	
TV use	9.06*** (0.01)	1.93 (0.01)	11.66*** (0.04)	3.73 (0.01)	3.44 (0.01)	8.16 (0.03)	6.91 (0.03)	2.41 (0.01)	36.31*** (0.09)	0.07
<2 hours (N = 7914)	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
vs.	8.92 (0.02)	1.98 (0.02)	12.07 (0.07)	3.81 (0.02)	3.44 (0.02)	8.23 (0.05)	7.06 (0.05)	2.48 (0.02)	37.08 (0.15)	
≥2 hours (N = 2762)	d = -0.14	d = 0.05	d = 0.11	d = 0.08	d = 0.00	d = 0.03	d = 0.06	d = 0.07	d = 0.10	
Video use	9.08*** (0.01)	1.91*** (0.01)	11.63*** (0.04)	3.75 (0.01)	3.44 (0.01)	8.15 (0.03)	6.88*** (0.03)	2.41 (0.01)	36.26*** (0.09)	0.11
< 2 hours	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
(N = 8521)	8.82 (0.02)	2.04 (0.02)	12.28 (0.08)	3.78 (0.03)	3.44 (0.02)	8.29 (0.06)	7.21 (0.05)	2.50 (0.03)	37.51 (0.18)	
vs.	d = -0.28	d = 0.14	d = 0.18	d = 0.03	d = 0.00	d = 0.05	d = 0.13	d = 0.08	d = 0.15	
≥2 hours (N = 2154)	9.06*** (0.01)	1.92*** (0.02)	11.64*** (0.04)	3.74 (0.01)	3.44 (0.01)	8.17 (0.03)	6.89*** (0.03)	2.41 (0.01)	36.28*** (0.09)	0.09
Video game use	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
<2 hours (N = 8464)	8.88 (0.02)	2.03 (0.02)	12.24 (0.08)	3.81 (0.03)	3.42 (0.02)	8.23 (0.06)	7.17 (0.05)	2.50 (0.03)	37.37 (0.18)	
vs.	d = -0.19	d = 0.08	d = 0.16	d = 0.06	d = -0.02	d = 0.02	d = 0.11	d = 0.08	d = 0.13	
≥2 hours (N = 2209)	9.03 (0.01)	1.94 (0.01)	11.75 (0.04)	3.75 (0.01)	3.44 (0.01)	8.18 (0.03)	6.94 (0.02)	2.43 (0.01)	36.49 (0.08)	0.09
Texting use	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
<2 hours	8.82 (0.06)	1.99 (0.06)	12.20 (0.22)	3.88 (0.07)	3.43 (0.05)	8.09 (0.15)	7.16 (0.14)	2.53 (0.07)	37.29 (0.48)	
(N = 10,385)	d = -0.21	d = 0.05	d = 0.11	d = 0.12	d = -0.01	d = -0.03	d = 0.10	d = 0.09	d = 0.10	
vs.	9.03*** (0.01)	1.94 (0.01)	11.75 (0.04)	3.75 (0.01)	3.44 (0.01)	8.18 (0.03)	6.94 (0.02)	2.43 (0.01)	36.49 (0.08)	0.12
≥2 hours (N = 291)	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
Social media use	8.74 (0.08)	1.98 (0.08)	12.49 (0.30)	3.84 (0.10)	3.59 (0.07)	8.21 (0.21)	7.19 (0.19)	2.31 (0.09)	37.62 (0.64)	
<2 hours	d = -0.28	d = 0.04	d = 0.19	d = 0.08	d = 0.16	d = 0.01	d = 0.11	d = -0.11	d = 0.14	
(N = 10,516)	9.03 (0.01)	1.94 (0.01)	11.75 (0.04)	3.75 (0.01)	3.44 (0.01)	8.19 (0.03)	6.94 (0.02)	2.43 (0.01)	36.50 (0.08)	0.09
Video chat use	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
< 2 hours	8.91 (0.07)	2.02 (0.07)	12.08 (0.25)	3.78 (0.08)	3.36 (0.06)	7.82 (0.18)	7.24 (0.16)	2.56 (0.08)	36.85 (0.54)	
(N = 10,448)	d = -0.12	d = 0.08	d = 0.08	d = 0.03	d = -0.08	d = -0.13	d = 0.14	d = 0.12	d = 0.04	
vs.	9.08*** (0.01)	1.91*** (0.01) vs.	11.62*** (0.04)	3.73 (0.01)	3.45 (0.01)	8.17 (0.03)	6.88*** (0.03)	2.41 (0.01)	36.24*** (0.09)	0.10
Non-TV screen use	vs.	2.00 (0.02)	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
<2 hours (N = 7914)	8.87 (0.02)	d = 0.09	12.18 (0.07)	3.81 (0.02)	3.42 (0.02)	8.22 (0.05)	7.15 (0.05)	2.49 (0.02)	37.27 (0.16)	
vs.	d = -0.22		d = 0.15	d = 0.08	d = -0.03	d = 0.02	d = 0.10	d = 0.08	d = 0.13	
≥2 hours (N = 2763)	9.16*** (0.02)	1.90 (0.02)	11.47*** (0.06)	3.71 (0.02)	3.45 (0.02)	8.14 (0.04)	6.84*** (0.04)	2.39 (0.02)	35.99*** (0.13)	0.08
Total screen use	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
<2 hours (N = 4110)	8.94 (0.01)	1.96 (0.01)	11.95 (0.05)	3.77 (0.02)	3.44 (0.01)	8.21 (0.03)	7.01 (0.03)	2.46 (0.02)	36.84 (0.10)	
vs.	d = -0.21	d = 0.06	d = 0.12	d = 0.04	d = -0.01	d = 0.03	d = 0.07	d = 0.05	d = 0.10	
≥2 hours (N = 6567)										

Estimated means after adjustment for study covariates. Bold cells indicate statistically significant differences (****P* < .001).

Table 3
Association between sleep-wake disturbances and time use of screen medias of screen medias on weekends.

Predictor	Sleep duration	Sleep onset latency	Insomnia	Sleep disordered breathing	Arousal disturbances	Sleep-wake transition disturbances	Excessive sleepiness	Sleep hyperhidrosis	Total sleep-wake disturbances	Average absolute <i>d</i>
	M (SE)	M (SE)	M (SE)	M (SE)	M (SE)	M (SE)	M (SE)	M (SE)	M (SE)	
TV use	9.06 (0.01)	1.91 (0.01)	11.59*** (0.05)	3.72*** (0.02)	3.43 (0.01)	8.13 (0.03)	6.87 (0.03)	2.38*** (0.02)	36.12*** (0.10)	0.07
<2 hours (N = 5941)	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
vs.	8.98 (0.02)	1.97 (0.01)	11.98 (0.05)	3.80 (0.02)	3.46 (0.01)	8.24 (0.04)	7.04 (0.04)	2.49 (0.02)	37.00 (0.12)	
≥2 hours (N = 4730)	<i>d</i> = −0.07	<i>d</i> = 0.08	<i>d</i> = 0.11	<i>d</i> = 0.05	<i>d</i> = 0.04	<i>d</i> = 0.04	<i>d</i> = 0.07	<i>d</i> = 0.08	<i>d</i> = 0.11	
Video use	9.10*** (0.01)	1.91*** (0.01)	11.58*** (0.04)	3.74 (0.01)	3.44 (0.01)	8.15 (0.03)	6.85*** (0.03)	2.40*** (0.01)	36.15*** (0.09)	0.12
<2 hours (N = 7708)	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
vs.	8.83 (0.02)	2.02 (0.02)	12.24 (0.07)	3.80 (0.02)	3.45 (0.02)	8.27 (0.05)	7.20 (0.05)	2.50 (0.02)	37.46 (0.15)	
≥2 hours (N = 2966)	<i>d</i> = −0.27	<i>d</i> = 0.11	<i>d</i> = 0.18	<i>d</i> = 0.06	<i>d</i> = 0.01	<i>d</i> = 0.04	<i>d</i> = 0.13	<i>d</i> = 0.10	<i>d</i> = 0.16	
Video game use	9.07*** (0.01)	1.90*** (0.01)	11.59*** (0.04)	3.73 (0.01)	3.44 (0.01)	8.13 (0.03)	6.87*** (0.03)	2.39*** (0.01)	36.16*** (0.10)	0.10
<2 hours (N = 7428)	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
vs.	8.93 (0.02)	2.02 (0.02)	12.16 (0.07)	3.79 (0.02)	3.44 (0.02)	8.29 (0.05)	7.12 (0.04)	2.51 (0.02)	37.31 (0.15)	
≥2 hours (N = 3243)	<i>d</i> = −0.14	<i>d</i> = 0.12	<i>d</i> = 0.15	<i>d</i> = 0.06	<i>d</i> = 0.00	<i>d</i> = 0.06	<i>d</i> = 0.10	<i>d</i> = 0.12	<i>d</i> = 0.13	
Texting use	9.03*** (0.01)	1.94 (0.01)	11.74 (0.04)	3.75 (0.01)	3.44 (0.01)	8.18 (0.03)	6.93 (0.02)	2.43 (0.01)	36.47 (0.08)	0.10
<2 hours (N = 10,264)	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
vs.	8.83 (0.05)	2.01 (0.05)	12.19 (0.19)	3.86 (0.06)	3.45 (0.05)	8.31 (0.13)	7.26 (0.12)	2.51 (0.06)	37.58 (0.40)	
≥2 hours (N = 410)	<i>d</i> = −0.20	<i>d</i> = 0.07	<i>d</i> = 0.11	<i>d</i> = 0.10	<i>d</i> = 0.01	<i>d</i> = 0.05	<i>d</i> = 0.15	<i>d</i> = 0.07	<i>d</i> = 0.14	
Social media use	9.03*** (0.03)	1.94 (0.01)	11.75 (0.04)	3.75 (0.01)	3.44 (0.01)	8.17 (0.03)	6.94 (0.02)	2.43 (0.01)	36.47*** (0.08)	0.14
<2 hours (N = 10,438)	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
vs.	8.75 (0.07)	2.06 (0.07)	12.48 (0.24)	3.79 (0.08)	3.59 (0.06)	8.59 (0.17)	7.35 (0.16)	2.48 (0.08)	38.27 (0.53)	
≥2 hours (N = 236)	<i>d</i> = −0.14	<i>d</i> = 0.11	<i>d</i> = 0.19	<i>d</i> = 0.04	<i>d</i> = 0.15	<i>d</i> = 0.15	<i>d</i> = 0.18	<i>d</i> = 0.04	<i>d</i> = 0.22	
Video chat use	9.03*** (0.01)	1.94 (0.01)	11.74 (0.04)	3.74 (0.01)	3.44 (0.01)	8.18 (0.03)	6.93 (0.02)	2.43 (0.01)	36.46 (0.08)	0.12
<2 hours (N = 10,322)	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
vs.	8.82 (0.06)	1.99 (0.05)	12.43 (0.20)	3.89 (0.07)	3.45 (0.05)	8.19 (0.14)	7.37 (0.13)	2.55 (0.06)	37.87 (0.44)	
≥2 hours (N = 344)	<i>d</i> = −0.20	<i>d</i> = 0.05	<i>d</i> = 0.18	<i>d</i> = 0.13	<i>d</i> = 0.01	<i>d</i> = 0.00	<i>d</i> = 0.20	<i>d</i> = 0.11	<i>d</i> = 0.17	
Non-TV screen use	9.09*** (0.01)	1.90*** (0.01)	11.55*** (0.05)	3.73 (0.02)	3.44 (0.01)	8.12 (0.03)	6.85*** (0.03)	2.40*** (0.01)	36.10*** (0.10)	0.10
<2 hours (N = 6821)	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
vs.	8.91 (0.02)	2.00 (0.02)	12.13 (0.06)	3.79 (0.02)	3.44 (0.02)	8.28 (0.04)	7.11 (0.04)	2.49 (0.02)	37.24 (0.13)	
≥2 hours (N = 3854)	<i>d</i> = −0.17	<i>d</i> = 0.10	<i>d</i> = 0.15	<i>d</i> = 0.04	<i>d</i> = 0.00	<i>d</i> = 0.06	<i>d</i> = 0.10	<i>d</i> = 0.09	<i>d</i> = 0.14	
Total screen use	9.15*** (0.02)	1.89 (0.02)	11.35*** (0.08)	3.68*** (0.03)	3.42 (0.02)	8.05 (0.05)	6.76*** (0.05)	2.36 (0.02)	35.62*** (0.17)	0.10
<2 hours (N = 2388)	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	vs.	
vs.	8.99 (0.01)	1.95 (0.01)	11.88 (0.04)	3.78 (0.01)	3.45 (0.01)	8.22 (0.03)	7.00 (0.03)	2.45 (0.01)	36.77 (0.09)	
≥2 hours (N = 8287)	<i>d</i> = −0.17	<i>d</i> = 0.06	<i>d</i> = 0.14	<i>d</i> = 0.08	<i>d</i> = 0.03	<i>d</i> = 0.07	<i>d</i> = 0.09	<i>d</i> = 0.10	<i>d</i> = 0.14	

Estimated means after adjustment for study covariates. Bold cells indicate statistically significant differences (****P* < .001).

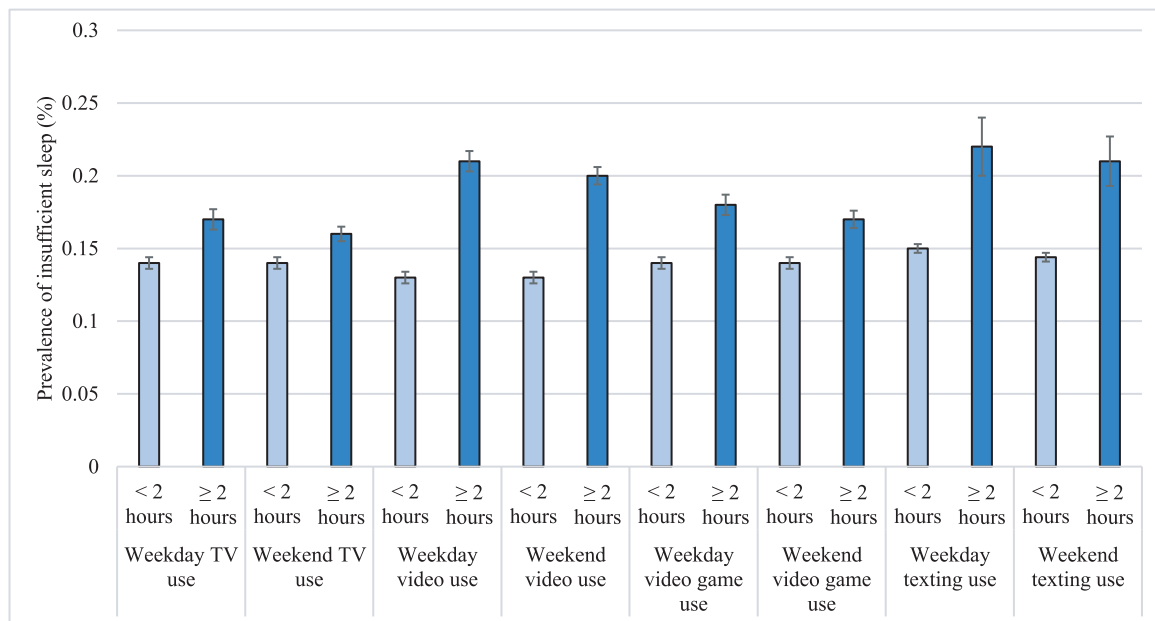


Fig. 1. Prevalence of insufficient sleep across ≥ 2 hours and < 2 hours of TV, video, video game, and texting use on weekdays and weekends. Adjusted for study covariates. Bars represent standard error of the mean.

children, whether these differences also extended to differences in the associations between screen use and sleep were explored. However, all interaction terms examining if Black children or Hispanic children had a differential association between screen media use and sleep than White children failed to reach significance (all P 's $> .001$).

Discussion

Greater TV, video, and video game use were the screen media types most frequently associated with decreased sleep duration, longer sleep onset latency, and increased sleep-wake disturbance symptom severity in 9 to 10-year-old children. These findings are partly surprising because prior research suggests that TV use may have weaker associations with sleep compared to more interactive types of screen uses such as social media.^{16,28,29} One reason use of these

types of screen media had the most associations with sleep-wake disturbances may be because these were the most prevalent kinds of screen media use within this sample of 9–10 year old children. Within the total sample, very few children (less than 4%) spent 2 or more hours a day texting, on social media, or video chatting. Moreover, the association between TV and video gaming use with sleep duration has been found to wane to non-significance from childhood to adolescence, whereas the association of portable electronic device use and sleep duration remains significant.¹⁶ As children get older, the associations between watching TV and videos with sleep may decrease as this time is replaced by or joined with cell phone use.

While use of TV, video, and video games had the most frequent significant associations with sleep, they did not necessarily have the largest magnitude of associations with sleep (social media use did). Of note is that TV use actually had the smallest average magnitude of

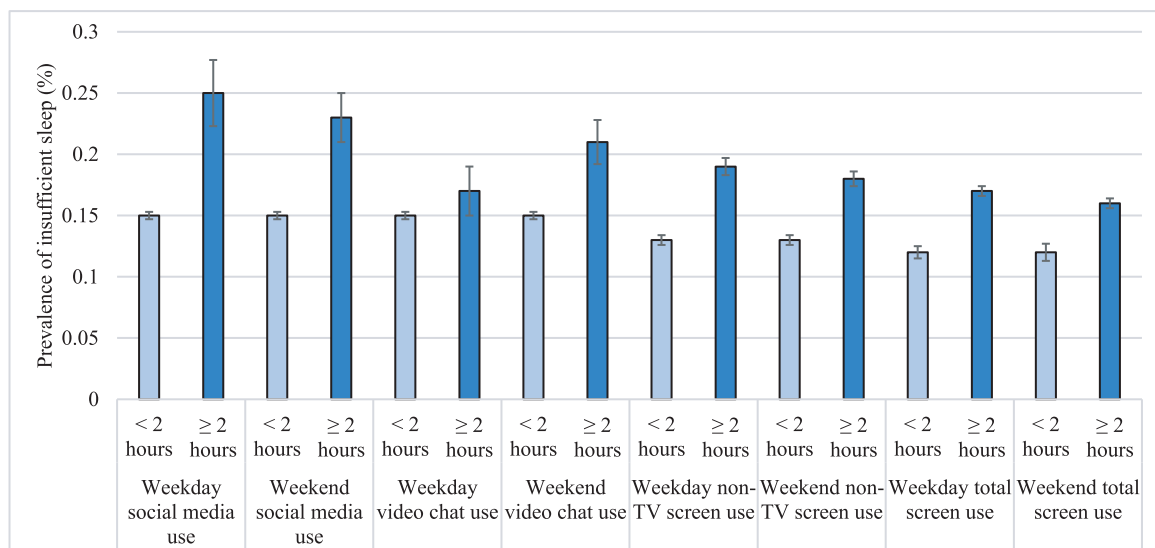


Fig. 2. Prevalence of insufficient sleep across ≥ 2 hours and < 2 hours of social media, video chat, total non-TV, and total screen use on weekdays and weekends. Adjusted for study covariates. Bars represent standard error of the mean.

Table 4

Relative risk of insufficient sleep between children with two hours or more versus children with less than two hours of screen media use.

Predictor	Relative risk of insufficient sleep
≥2 hours vs. <2 hours of weekday TV use	1.27***
≥2 hours vs. <2 hours of weekend TV use	1.23***
≥2 hours vs. <2 hours of weekday video use	1.59***
≥2 hours vs. <2 hours of weekend video use	1.60***
≥2 hours vs. <2 hours of weekday video game use	1.36***
≥2 hours vs. <2 hours of weekend video game use	1.33***
≥2 hours vs. <2 hours of weekday texting use	1.44
≥2 hours vs. <2 hours of weekend texting use	1.39
≥2 hours vs. <2 hours of weekday social media use	1.56
≥2 hours vs. <2 hours of weekend social media use	1.50
≥2 hours vs. <2 hours of weekday video chat use	1.13
≥2 hours vs. <2 hours of weekend video chat use	1.42
≥2 hours vs. <2 hours of weekday non-TV screen use	1.42***
≥2 hours vs. <2 hours of weekend non-TV screen use	1.43***
≥2 hours vs. <2 hours of weekday total screen use	1.55***
≥2 hours vs. <2 hours of weekend total screen use	1.47***

Adjusted for study covariates.

*** $P < .001$.

association with sleep. Considering that most prior research was conducted before the advent of more portable and interactive screen media types (e.g., smartphones) and have mostly reported findings with TV use, these prior studies do not speak to the broader relations between different types of screen media use and sleep-wake disturbances in children.^{7–10} Moreover, social media use had a noticeably large association with sleep outcomes relative to the other screen media uses, though these associations rarely achieved statistical significance, perhaps due to the low rates of use in this sample. The larger associations with social media use converge with other

studies.^{28,30} Given that social media use involves using a screen, actively browsing a social media website, processing social information about others, and potentially having positive and negative interactions with other people, these larger effects are not particularly surprising. Future research should continue examining other kinds of screen media use in addition to TV use and keep in mind that the relevance of particular screen medias may wax and wane across the lifespan.

While specific kinds of screen media use consistently related to sleep-wake disturbance symptom severity, specific sleep-wake disturbances were consistently associated with screen media use. In particular, screen media uses were associated with sleep-wake disturbances characterized by difficulties falling and staying asleep and excessive daytime sleepiness rather than disturbance characterized by breathing, arousal, sleep-wake transition, and hyperhidrosis disturbances. This supports prior findings with TV use and extends them to additional types of screen medias.^{3,9,10} Assuming that screen media use undermines sleep, it is not surprising that greater use of screen media would be associated with exacerbated insomnia and excessive sleepiness symptoms because insomnia and sleepiness can originate from insufficient sleep and difficulties falling asleep. In line with this, multiple screen media were associated with longer sleep onset latency and shorter sleep duration. A variety of nonmutually exclusive processes may explain these associations. Spending too much time on screen media may displace sleep, the blue light emitted from screens may delay melatonin production and delay sleep, and content on screen media may be arousing and inhibit sleep.^{4,5} However, it is also possible that people engage in greater screen media use if they have difficulties sleeping and when they feel too lethargic to engage in more active activities. Given that both possibilities are plausible, longitudinal designs will be needed to better understand the directionality of these associations.

Another interesting pattern was that typical weekend (compared to weekday) screen media uses appeared to be more closely tied to sleep-wake disturbance symptom severity, though more targeted research on this possibility is needed. This finding is reminiscent of Li and colleagues' finding that child TV viewing on the weekends had a particularly strong association with sleep timing and duration.³ One explanation is that there are more social constraints and influences on screen media use and sleep on the weekdays (e.g., school schedules) than weekends, which may limit their association. Additionally,

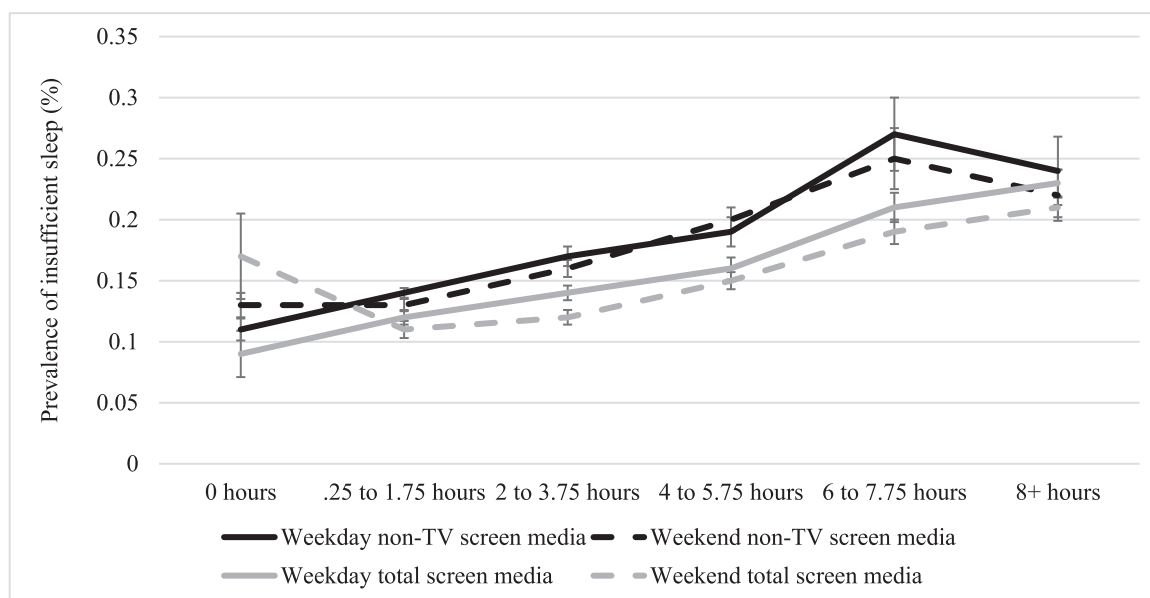


Fig. 3. Prevalence of insufficient sleep across continuous screen media use categories. Adjusted for study covariates. Bars represent standard error of the mean.

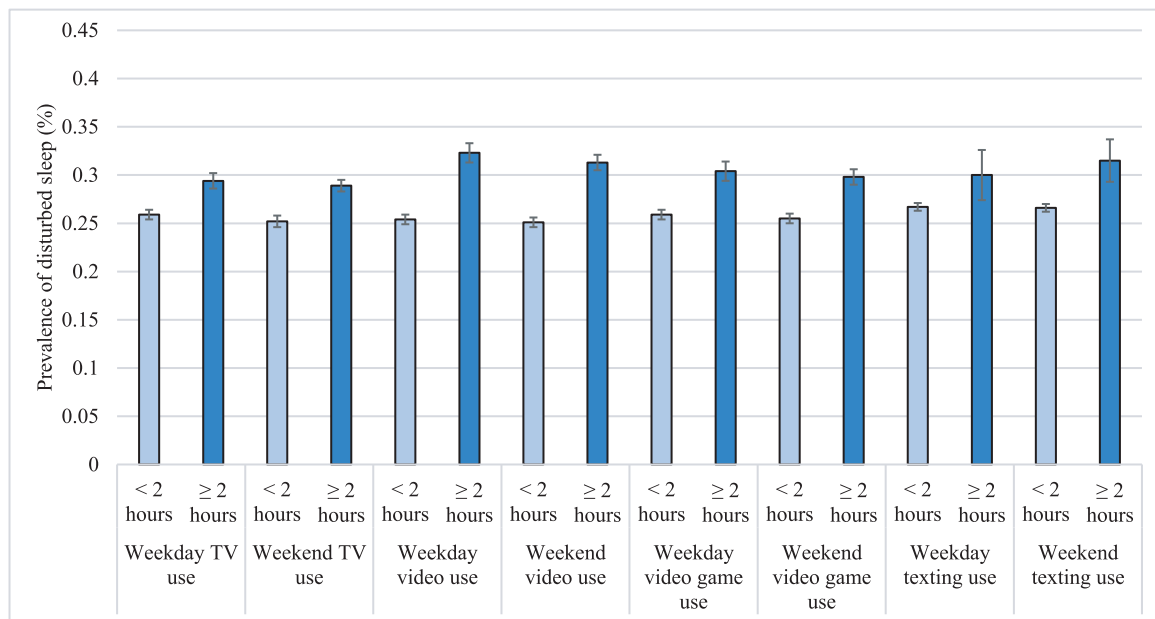


Fig. 4. Prevalence of disturbed sleep across ≥ 2 hours and < 2 hours of TV, video, video game, and texting use on weekdays and weekends. Adjusted for study covariates. Bars represent standard error of the mean.

parental regulations on screen media use may be stronger on school-days than weekends.³¹ In the current study, more children engaged in 2+ hours of screen media during the weekend and supplemental analyses demonstrated that they also spend more time on average on screen media during the weekend than weekdays (see supplemental Table 1). This greater use may translate to a greater impact on sleep, assuming that screen media are influencing sleep rather than vice versa.

It is important to note that greater TV, video, and video game use on both weekdays and weekends was associated with a greater risk of having clinically relevant insufficient sleep and disturbed sleep. Additionally, each one hour increase in both non-TV screen media use and total screen media use predicted increased odds of having insufficient or disturbed sleep. These findings suggest that not only is greater use of screen media associated with worse sleep, but that

these associations may warrant clinical attention. Assuming that these associations are the product of screen use disturbing sleep, these findings highlight that regulating screen use may be one way to improve sleep in childhood and adolescence. Such regulations could be implemented individually with the child as well as within the family as parents play a protective regulatory role in screen media use as well as provide role models for its use.^{4,31}

Finally, ethnoracial differences emerged between White and Black children wherein Black children reported greater use of all types of screen media and also reported greater sleep disturbances. These findings are in line with previous work showing that Black children are more likely to have a TV in their bedroom and have short actigraphic and self-reported sleep.^{17–19} In addition, Hispanic/Latinx children had better sleep than White children, a finding reminiscent of the Hispanic health paradox wherein Hispanic/Latinx individuals actually have better

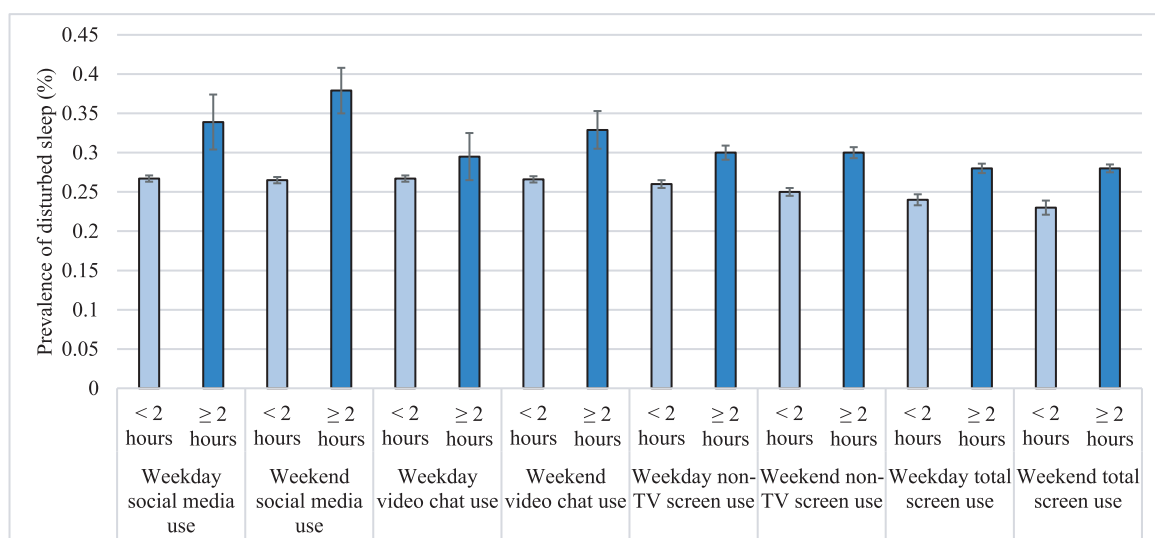


Fig. 5. Prevalence of disturbed sleep across ≥ 2 hours and < 2 hours of social media, video chat, total non-TV, and total screen use on weekdays and weekend. Adjusted for study covariates. Bars represent standard error of the mean.

Table 5

Relative risk of having disturbed sleep between children with 2 hours or more and children with less than two hours of screen media use.

Variable	Relative risk
≥2 hours vs. <2 hours of weekday tv use	1.14***
≥2 hours vs. <2 hours of weekend tv use	1.15***
≥2 hours vs. <2 hours of weekday video use	1.27***
≥2 hours vs. <2 hours of weekend video use	1.25***
≥2 hours vs. <2 hours of weekday video game use	1.18***
≥2 hours vs. <2 hours of weekend video game use	1.17***
≥2 hours vs. <2 hours of weekday texting use	1.12
≥2 hours vs. <2 hours of weekend texting use	1.18
≥2 hours vs. <2 hours of weekday social media use	1.26
≥2 hours vs. <2 hours of weekend social media use	1.42***
≥2 hours vs. <2 hours of weekday video chat use	1.10
≥2 hours vs. <2 hours of weekend video chat use	1.23
≥2 hours vs. <2 hours of weekday non-TV screen use	1.17***
≥2 hours vs. <2 hours of weekend non-TV screen use	1.18***
≥2 hours vs. <2 hours of weekday total screen use	1.17***
≥2 hours vs. <2 hours of weekend total screen use	1.20***

Adjusted for study covariates.

*** $P < .001$.

health outcomes in comparison to White individuals despite belonging to a minority group within the United States.³² The differences in sleep outcomes, however, were not accounted for by differences in screen use and thus other factors seem to be at play in the sleep disparities. Moreover, differential relations among screen uses and sleep outcomes did not emerge in moderation analyses. Altogether, these findings indicate that future research should consider demographic differences in screen use and sleep and seek understand why such differences emerge.

Limitations

There are several limitations to note for this study. First, the findings are cross-sectional, which obscures understanding the direction of the effects. It is plausible that greater screen media use may disturb

sleep and exacerbate sleep-wake disturbance symptom severity, that people with disturbed or insufficient sleep may engage in more screen media use, or both. Longitudinal and experimental designs would be valuable assets in determining the direction of the effects as well as controlling for third-variables that may confound findings.

Second, sleep or screen media use were both person-reported and therefore are subject to reporting biases and limitations. It is important to note that screen media use was child-reported while sleep was parent-reported. On one hand, using parent-reported instead of child-reported sleep should avoid some common method biases (e.g., negative reporting styles) which inflate estimated relations between screen use and sleep; on the other hand, parents may not have full or completely accurate knowledge of their child's sleep behavior. As always, obtaining behavioral measurements of screen media use and sleep would increase confidence in results, though obtaining such measurements in a large sample of children would be difficult and costly. It is also important to note that screen and sleep measures were collected throughout the year, and when the data were collected (e.g., during the school year vs. summer break) can influence screen use and sleep and may change how they relate.

Relatedly, data from this study did not capture timing of screen media use. Screen media before bedtime can exert a stronger influence on sleep than screen media used earlier in the day.³³ Screen media used in the first half of the day may have little to no influence on the next night's sleep and dilute associations between screen media use and sleep. Assessments in this study also did not assess what screen media was being used for (e.g., recreational vs. academic uses) nor explicitly assess what kind of device the screen media was used on (e.g., watching TV on a traditional TV vs. watching TV on a cell phone). Screen media used for recreational purposes may be particularly impactful on sleep as recreational media use may be more likely to produce arousing positive affect (e.g., excitement) that interferes with sleep. Additionally, screen media used on portable devices, which are held closer to the face resulting in a stronger alerting effect of blue light, may have a stronger association with

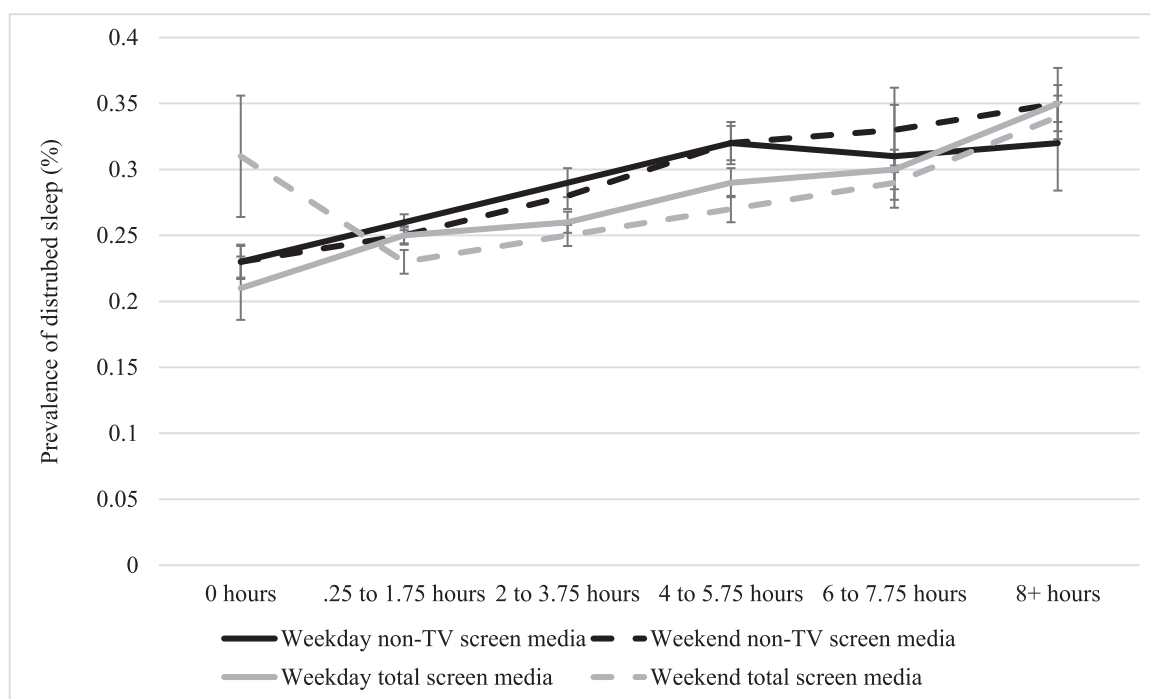


Fig. 6. Prevalence of disturbed sleep across continuous screen media use categories. Adjusted for study covariates. Bars represent standard error of the mean.

Table 6
Comparison of mean differences in screen use and sleep between White and other ethn racial groups.

	White children(N = 5922)	Black children(N = 1840)	Asian children(N = 543)	Indigenous-American children(N = 0.254)	Hispanic/Latinx children (N = 2062)
Outcome	M (SE)	M (SE)	M (SE)	M (SE)	M (SE)
Weekday TV use	0.24 (0.01)	0.35 (0.01)***	0.20 (0.02)	0.30 (0.03)	0.25 (0.01)
Weekend TV use	0.43 (0.01)	0.51 (0.01)***	0.38 (0.02)	0.46 (0.03)	0.44 (0.01)
Weekday video use	0.17 (0.01)	0.30 (0.01)***	0.16 (0.02)	0.25 (0.02)	0.21 (0.01)***
Weekend video use	0.24 (0.01)	.38 (0.01)***	0.23 (0.02)	0.33 (0.03)	0.30 (0.01)***
Weekday video game use	0.19 (0.01)	0.28 (0.01)***	0.18 (0.02)	0.24 (0.02)	0.19 (0.01)
Weekend video game use	0.29 (0.01)	0.39 (0.01)***	0.28 (0.02)	0.36 (0.03)	0.28 (0.01)
Weekday texting use	0.02 (0.00)	0.07 (0.00)***	0.02 (0.01)	0.03 (0.01)	0.03 (0.00)
Weekend texting use	0.03 (0.00)	0.09 (0.01)***	0.02 (0.01)	0.04 (0.01)	0.04 (0.00)
Weekday social media use	0.01 (0.00)	0.04 (0.00)***	0.01 (0.01)	0.02 (0.01)	0.00 (0.00)
Weekend social media use	0.02 (0.00)	0.06 (0.00)***	0.01 (0.01)	.02 (0.01)	0.01 (0.00)
Weekday video chat use	0.01 (0.00)	0.05 (0.00)***	0.01 (0.01)	.01 (0.01)	0.02 (0.00)
Weekend video chat use	0.02 (0.00)	0.08 (0.00)***	0.02 (0.01)	0.04 (0.01)	0.03 (0.00)
Sleep duration	9.19 (0.01)	8.56 (0.03)***	9.13 (0.04)	8.76 (0.06)***	9.96 (0.02)***
Sleep onset latency	1.96 (0.01)	2.01 (0.03)	1.86 (0.04)	1.99 (0.06)	1.83 (0.02)***
Insomnia	11.81 (0.05)	12.08 (0.09)	11.60 (0.16)	12.47 (0.23)	11.34 (0.09)***
Sleep disordered breathing	3.66 (0.02)	4.04 (0.03)***	3.85*** (0.03)	3.81 (0.08)	3.76 (0.03)
Arousal dist.	3.46 (0.02)	3.37 (0.02)	3.50 (0.04)	3.57 (0.06)	3.41 (0.02)
Sleep-wake transition dist.	8.25 (0.04)	7.86 (0.07)***	3.30 (0.11)	8.43 (0.16)	8.20 (0.06)
Excessive sleepiness	6.96 (0.03)	6.99 (0.06)	6.91 (0.10)	7.34 (0.10)	6.83 (0.06)
Sleep hyperhidrosis	2.41 (0.02)	2.44 (0.03)	2.52 (0.05)	2.66 (0.07)	2.43 (0.03)
Total sleep-wake dist.	36.55 (0.11)	36.77 (0.20)	36.68 (0.35)	38.28 (0.51)	35.96 (0.18)

Dist. = Disturbances. Estimated means after adjustment for covariates. Bold cells indicate statistically significant differences in comparison to White children (*** $P < .001$).

sleep.¹⁶ Future research incorporating timing, purpose of use, and kind of device use would be able to account for and examine such possibilities.

Finally, this study was predominately composed of children from White and higher socioeconomic status backgrounds. Given known disparities in screen use and sleep across socioeconomic and racial/ethnic backgrounds, which our findings replicate for Black children, additional research sampling participants from more diverse backgrounds is needed to better understand the interplay between screen use and sleep across demographic factors.^{17–19}

Conclusion

Greater screen media use 9- to 10- year-old children was not only associated with decreased sleep duration and increased sleep onset latency, but also greater symptom severity of different kinds of sleep-wake disturbances, particularly insomnia and excessive sleepiness. TV, video, and video game use greater than two hours were especially associated with increased severity of multiple kinds of sleep-wake disturbances. Although rates of social media use were low in this sample, estimates of effect size suggest that too much time on social media may be particularly associated with poor sleep. In sum, these findings extend prior work that too much time on screen media is associated with poor sleep by demonstrating that greater use of various types of screen media are associated with increased severity of multiple types of sleep-wake disturbances.

Conflict of interest

The author has no conflicts of interest to disclose.

Acknowledgments

Data used in the preparation of this article were obtained from the Adolescent Brain Cognitive Development (ABCD) Study (<https://abcdstudy.org>), held in the NIMH Data Archive (NDA). This is a multi-site, longitudinal study designed to recruit more than 10,000 children age 9–10 and follow them over 10 years into early adulthood. The

ABCD Study is supported by the National Institutes of Health and additional federal partners under award numbers U01DA041048, U01DA050989, U01DA051016, U01DA041022, U01DA051018, U01DA051037, U01DA050987, U01DA041174, U01DA041106, U01DA041117, U01DA041028, U01DA041134, U01DA050988, U01DA051039, U01DA041156, U01DA041025, U01DA041120, U01DA051038, U01DA041148, U01DA041093, U01DA041089, U24DA041123, U24DA041147. A full list of supporters is available at <https://abcdstudy.org/federal-partners.html>. A listing of participating sites and a complete listing of the study investigators can be found at <https://abcdstudy.org/principal-investigators/>. ABCD consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in analysis or writing of this report. This manuscript reflects the views of the authors and may not reflect the opinions or views of the NIH or ABCD consortium investigators. The ABCD data repository grows and changes over time. The ABCD data used in this report came from DOI 10.15154/1503209.

Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.sleh.2020.07.002.

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